Name: [redacted]

PCS125 QUIZ – Second Week of Tutorial #1 Student #: [redacted]

A simple pendulum has a bob of mass of 0.250 kg and a length of 0.5 m. It is displaced through an angle of 15° and then released. Calculate the following. (Give your answers with three digits)

(a) Determine the angular frequency, ω.

(b) What is the maximum angular speed of the pendulum and the maximum speed of the bob, v_{max}?

(c) What is the maximum angular acceleration of the bob and the maximum linear acceleration of the bob, a_{max}?

(d) What is the maximum restoring force of the bob, F_{max}?

(e) What is the total energy, E, of the pendulum bob when the angular displacement is 5.0°?

(f) If 10 seconds after starting the motion, the amplitude is 12°, what is the damping parameter, B?

a) $\omega = \sqrt{\frac{g}{L}} = \sqrt{\frac{9.81}{0.5}} = \boxed{4.43 \ \frac{rad}{s}}$

$\frac{5}{6}$

b) $v_{angular} = \omega A = 4.429 \ \frac{rad}{s} \times \frac{15 \times \pi}{180} = \boxed{1.16 \ \frac{rad}{s}}$

$v_{linear} = v_{angular} \times l = 1.160 \ \frac{rad}{s} \times 0.5 m = \boxed{0.580 \ \frac{m}{s}}$

c) $a_{angular} = \omega^2 A = (4.429 \ \frac{rad}{s})^2 \times \frac{15 \times \pi}{180} = \boxed{5.14 \ \frac{rad}{s^2}}$

$a_{linear} = a_{angular} \times l = 5.135 \ \frac{rad}{s} \times 0.5 m = \boxed{2.57 \ \frac{m}{s^2}}$

d) $F_{max} = m \times a_{max}$

$= 0.250 Kg \times 5.135 \ \frac{rad}{s^2}$

$= \boxed{1.28 \ N}$

e) $E_T = \frac{1}{2} m v_{max}^2$

$= \frac{1}{2} (0.250 Kg) (1.160 \ \frac{rad}{s})^2$

$= \boxed{0.168 \ J}$

f) $A = A_0 e^{-bt}$

$12 = 15 e^{-b(10s)}$

$\frac{4}{5} = e^{-10b}$

$\ln \frac{4}{5} = -10b$

$b = \boxed{0.0223 s^{-1}}$

Name: [redacted]

PCS125 QUIZ – Second Week of Tutorial #2

Student #: [redacted]

a) The pressure in a sound wave traveling in air (ρ=1.29 kg/m^3) is given by the equation

$$P=1.9\sin(\pi(0.6x-310t))$$

where x is in metres and P in Pascal (Pa). Determine the pressure at x=1.3m and t=0.2 s, the wavelength, the speed of the wave and the displacement amplitude.

b) Two sound waves give rise to pressure variations at a certain point in space given by

$$p_1 = p_m \sin(\omega t)$$
$$p_2 = p_m \sin(\omega t - \phi)$$

If p_m=1.7 Pa and f=420 Hz, determine the pressure displacement of the resultant wave at that point in space at t=1.5s for the two values of phase difference: $\phi_1 = \pi/9$, $\phi_2 = \pi/6$.

$\frac{8}{8}$

a) $P = 1.9 \sin(\pi 0.6 (1.3\,m) - \pi 310 (0.2\,s))$

$= \boxed{1.21\ Pa}$

$k = \frac{2\pi}{\lambda} \Rightarrow 0.6\pi = \frac{2\pi}{\lambda} \Rightarrow \lambda = \frac{2\pi}{0.6\pi} \Rightarrow \boxed{\lambda = 3.33\ m}$

$w = 2\pi f \Rightarrow f = \frac{w}{2\pi} \Rightarrow f = \frac{\pi 310}{2\pi} \Rightarrow f = 155\ Hz$

$v = \lambda f \Rightarrow v = \frac{10}{3} \times 155 = \boxed{516.7\ \frac{m}{s}}$

$S_{max} = \frac{P_{max}}{\rho v w} = \frac{1.9}{1.29 \times 516.7 \frac{m}{s} \times 310\pi} = \boxed{2.93 \times 10^{-6}\ m}$

b) $\phi_1 = \frac{\pi}{9}$

$P_1 = 1.7 \sin(2\pi \times 420 \times 1.5)$

$P_2 = 1.7 \sin(2\pi \times 420 \times 1.5 - \frac{\pi}{9})$

$P_1 + P_2 = -0.58\ pa$

$\Delta P = \boxed{-0.58\ pa}$

$\phi_2 = \frac{\pi}{6}$

$P_2 = 1.7 \sin(2\pi \times 420 \times 1.5 - \frac{\pi}{6})$

$P_1 + P_2 = -0.85\ pa$

$\Delta P = \boxed{-0.85\ pa}$

Name: [redacted]

PCS125 QUIZ – Second Week of Tutorial #2 **Student #:** [redacted]

A system consists of four particles, each of mass m=22.00 g, located at the corners of a square with sides of length l=30.0 cm each.

(a) What is the total potential energy, U, of the system?

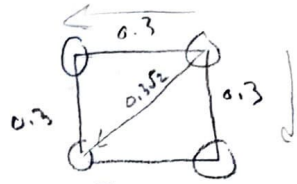
(b) How much work, W, is required to move one particle from the set?

(c) How much energy, E, is required to remove two particles at opposing corners?

(d) How much energy, E, is required to remove to adjacent particles?

$\sqrt{2}\,0.3$

$0.3\sqrt{2}$



a) $U = -G \times 10^{-6}\left[\frac{484\,g}{0.3\,m} + \frac{484\,g}{0.3\,m} + \frac{484\,g}{0.3\sqrt{2}\,m} + \frac{484\,g}{0.3\sqrt{2}\,m} + \frac{484}{0.3\,m} + \frac{484}{0.3\,m}\right]$

$= -5.83 \times 10^{-13}$ J

8/8

b) $U = G \times 10^{-6}\left[\frac{484\,g}{0.3\,m} + \frac{484\,g}{0.3\,m} + \frac{484\,g}{0.3\sqrt{2}\,m}\right]$

$= 2.91 \times 10^{-13}$ J

c) $U = G \times 10^{-6}\left[\frac{484\,g}{0.3\,m} \times 4 + \frac{484\,g}{0.3\sqrt{2}\,m}\right]$

$= 5.065 \times 10^{-13}$ J

d) $U = G \times 10^{-6}\left[\frac{484\,g}{0.3\,m} + \frac{484\,g}{0.3\,m} + \frac{484\,g}{0.3\sqrt{2}\,m} + \frac{484\,g}{0.3\,m} + \frac{484\,g}{0.3\sqrt{2}\,m}\right]$

$= 4.75 \times 10^{-13}$ J

Name: [redacted]

PCS125 QUIZ – Second Week of Tutorial #4 Student #: [redacted]

A particle of mass 3 mg and charge q = 2.4 mC moves parallel to the x-axis with an initial speed of 1.82×10^2 m/s at a point of coordinate x_1 = 1.0cm. Its speed is reduced to 0.48×10^2 m/s at a point of coordinate x_2 = 5.0cm.

Calculate the electric potential difference V_1 - V_2. Show all your work.

At which point does the particle have a higher potential energy?

4.5/6

$$V_1 - V_2 = \frac{K_i - K_f}{q}$$

$$= \frac{1}{2.4\times10^{-3}C}\left[\frac{1}{2}\times3\times10^{-6}Kg\times(1.82\times10^{2}\frac{m}{s})^2 - \frac{1}{2}\times3\times10^{-6}Kg\times(0.48\times10^{2}\frac{m}{s})^2\right]$$

$$= \frac{0.04623J}{2.4\times10^{-3}C}$$

$$= 19.2625\ V \simeq \boxed{19V}$$ −0.5

$\because V_1 - V_2 > 0$

the voltage is higher at 1.0 cm

−1

energy not voltage